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Electronic Practical Work

Microcontrollers Security

Writing In Memory

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A security issue is always a problem of reading or writing in the memory. A microcontroller has a Flash memory where programs are stored and an SRAM memory where to store variables and various data. Some microcontrollers also integrate an EEPROM.
The procedure for reading in FLASH or SRAM is similar. On the other hand, writing is completely different because of the technologies that do not allow the same granularity in writing.
work to be provided

- What do the addresses 0x8000 0000, 0x0808 0000 and 0x2000 0000 correspond to?
- Declare an unsigned 32-bit data variable.
- Read the data at addresses 0x0800 0000 and 0x2000 0000 and save this value in variables. Give the value read each time.
- Write 0xDEADBEEF to address 0x2000 0000 and read it back with the data to confirm that everything is running smoothly.
- Try the same manipulation with the addresses in the Flash
- Read the values in the FLASH memory starting at address 0x0802 0000. What values are saved? With these values, is the memory erased?
- Following the procedure 'Read from bank 1 while programming bank 2' explained in the reference manual, write 0xDEADBEEF to address 0x08080 0000 and 0xAABBCCDD to address 0x8080 0004. Don't forget that the FLASH must be unlocked before you can write to the FLASH.
- Change the value 0xAABBCCDD to 0x01234567 and check that you cannot write a new value in the flash memory. Explain why.
- Do a total Erase of Bank2 and check that the memory bytes are equal to 0xFF again…To do this, follow the procedure 'Read from Bank1 while page erasing in bank2'. Indeed, this procedure allows you to execute a program in bank1 while bank2 is erased.
- Check that you can write to the Flash again by repeating the previous operations

Writing in the flash memory consumes a high current. To detect if a program performs unexpected operations with the flash, you just need to check the current consumption.

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